Constructor University Bremen

MPFR LAB II

Part 1: Linear Control System MATLAB and Simulink

Spring Semester 2025

Task1.1 - Task1.3

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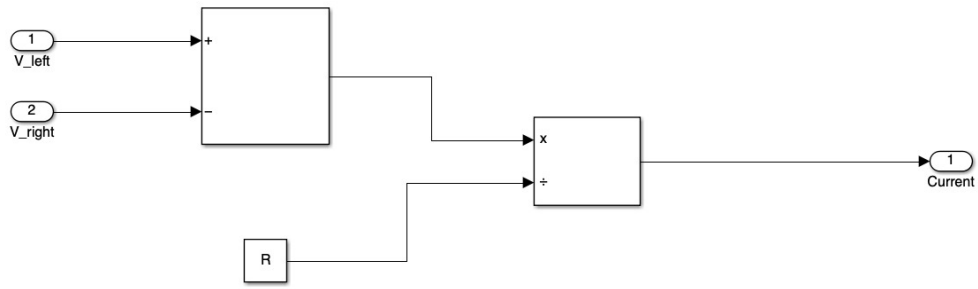
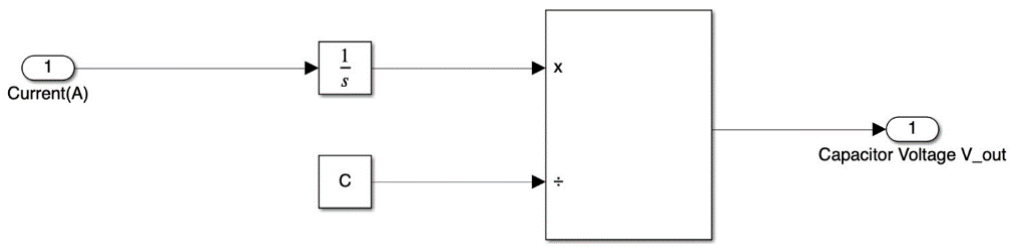
Task 1.1

Explain why Fig 1.4 represents the model of a capacitor by writing down the equation it implements





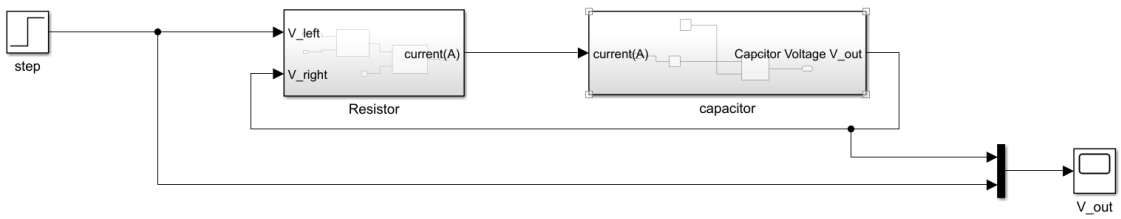

```
Matlab command-line:
>> R = 1e6
R =
      1000000

>> C = 1e-6
C =
      1.0000e-06
```

Equation:

$$V = \frac{1}{C} \times \int I$$

$$I = C \times \frac{dV}{dt}$$

Task1.2



What is the time-constant of this system?

Time-constant of this system = RC

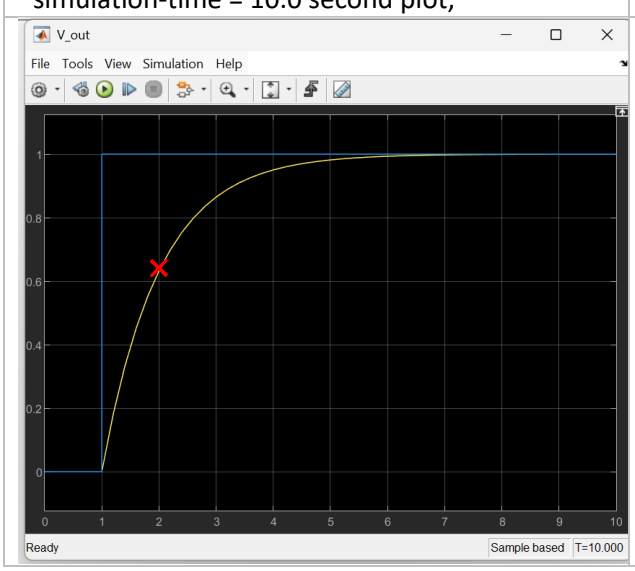
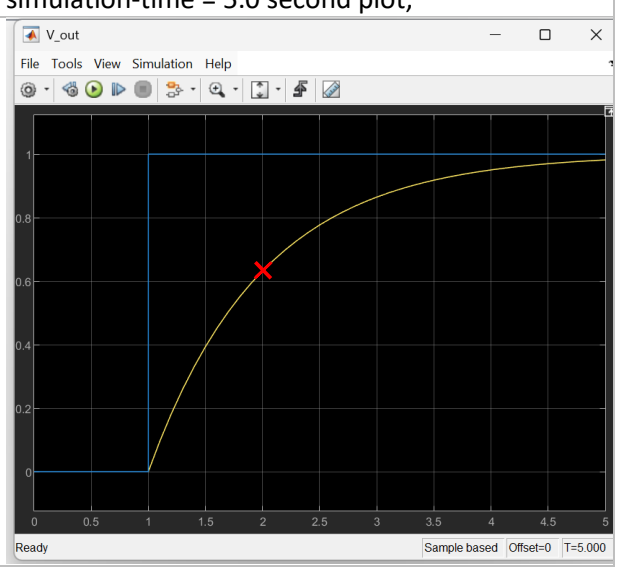
= (1e6)(1e-6)

= 1 second

How can you see it in the plot? Change the simulation-time from the default 10.0 seconds to 5 times this time constant.

Run the system by pressing Ctrl+T or by clicking the run-button. Look at the output by double clicking the scope.

The time-constant of the system can be observed in the plot by the time required for the capacitor to charge from 0% to 63% of the voltage value.

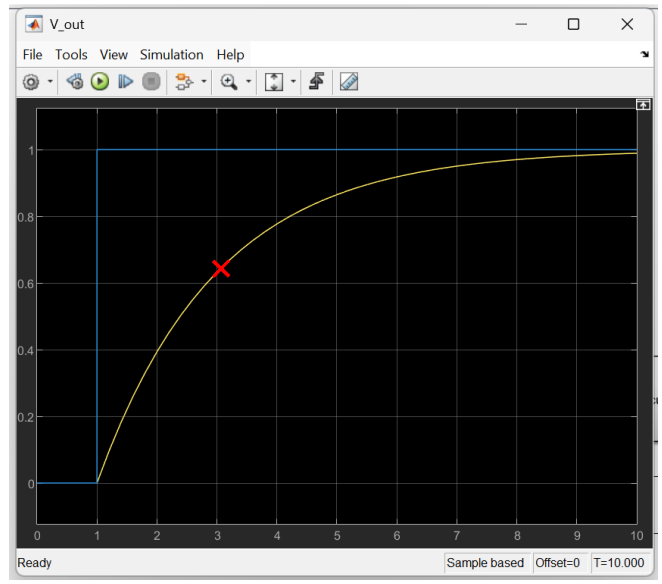
simulation-time = 10.0 second plot,	simulation-time = 5.0 second plot,
	

From both plot, the time-constant is 1 second.

Now change the variables R and C in the workspace and rerun the simulation. Does the scope display change as expected?

R is changed from 1e6 to 2e6.

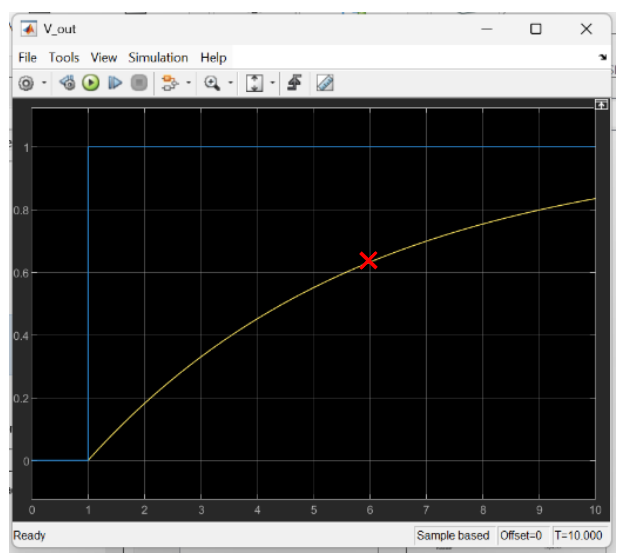
```
Matlab command-line:
>> R = 2e6
  R =
        2000000

  >> C = 1e-6
  C =
        1.0000e-06
```



Scope display changed time constant from 1 second to 2 seconds.

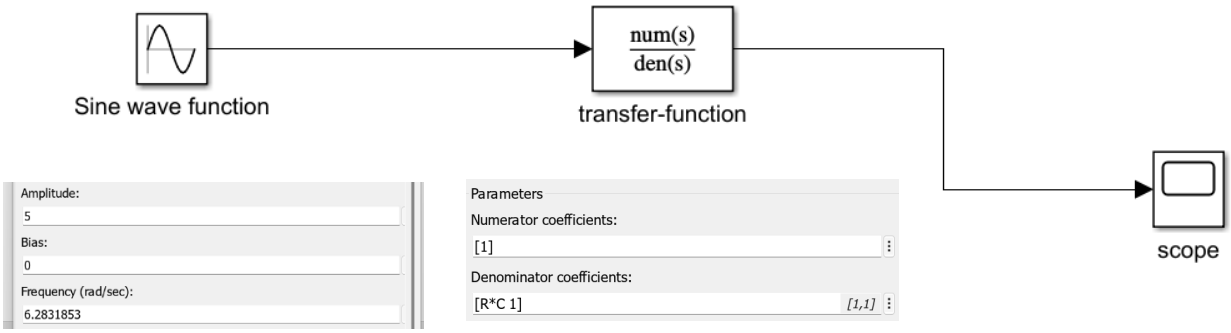
C is changed from 1e-6 to 5e-6.

```
Matlab command-line:
>> R = 1e6
  R =
        1000000

  >> C = 5e-6
  C =
        5.0000e-06
```



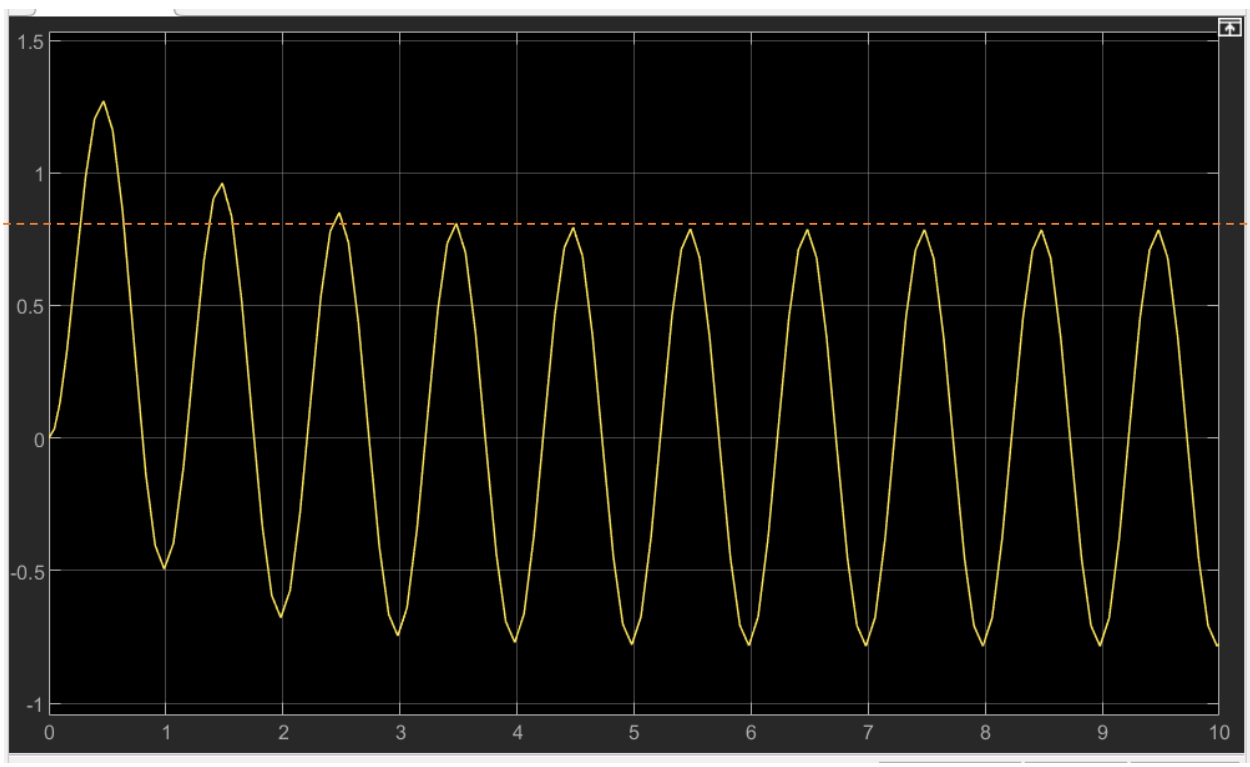
Scope display changed time constant from 1 second to 5 seconds.

Task 1.3



1 Hz = 2π rad/s R = 1e6

= 6.2831853 rad/s C = 1e-6



1. How many seconds does it take for the initial transient to die off in the output response?

Around 3.5 seconds, it began to stabilize.

2. What is the expected gain-ratio (ratio of output to input amplitudes) from theory? You can find this by replacing s by jω in the transfer-function and evaluating |H (jω)|.

the expected gain ratio:

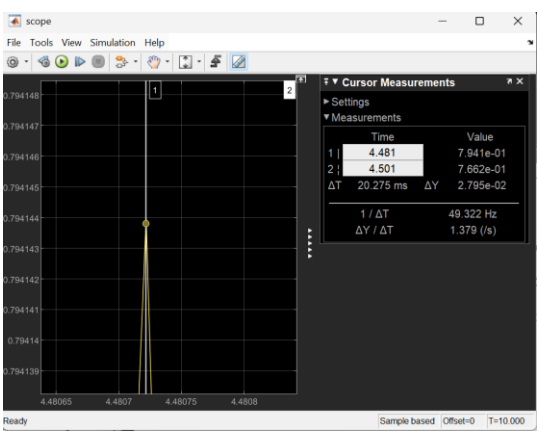
$$H(s) = \frac{1}{RCs + 1}$$

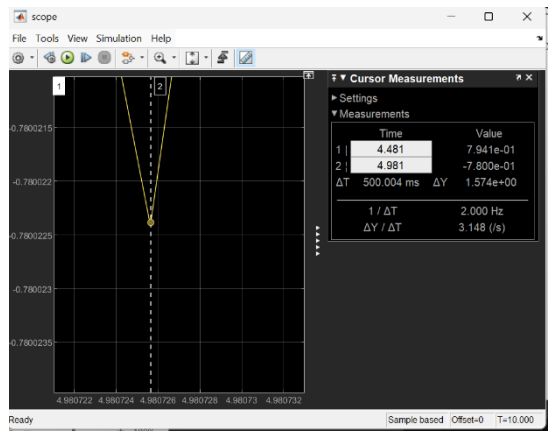
$$H(j\omega) = \frac{1}{RCj\omega + 1}$$

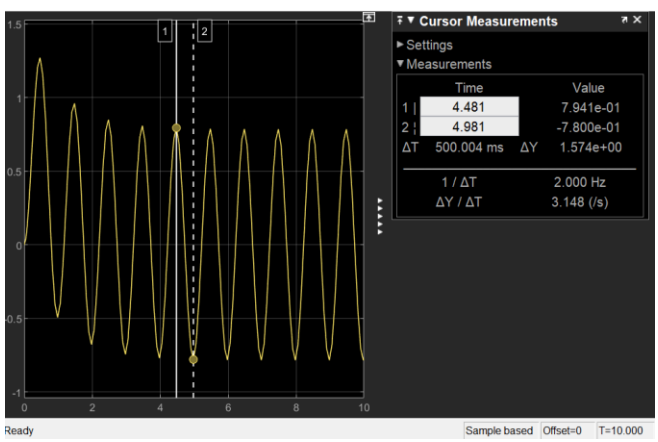
Since, R*C = 1 and ω = 2πrad/s.

$$H(j\omega) = \frac{1}{2\pi j + 1}$$

3. Zoom in the scope to find the amplitude ratio of the output wave to the input wave. Is it as expected?







Zoomed in to find the difference between the crest and trough which is 1.574.

Therefore, the amplitude is around 1.574/2 = 0.787.

It is as expected, because when you divide this amplitude by the input 5V is 0.787/5 = 0.1574.